Bloch-McConnell Simulation Study – Comparison of Various CEST Simulation Frameworks

Markus Huemer[1], Patrick Schuenke[2,3], Rudolf Stollberger[1], Viktoria Buchegger[1], Martin Juschitz[1], Nick Scholand[1], Moritz Simon Fabian[4], Felix F. Zimmermann[2], Christoph Kolbitsch[2], Nikita Vladimirov[5], Or Perlman[5,6], Swee Qi Pan[7,8], Yee Kai Tee[7], Ouri Cohen[9], Huabin Zhang[10], Jianpan Huang[10], Jian Hou[11], Qing Zeng[12,13], Nirbhay Yadav[12,13], Jiadi Xu[12,13], Kexin Wang[12,13,14], Beomgu Kang[12,13], Munendra Singh[12,13], Hye-Young Heo[12,13], Jingpu Wu[13,15], Jinyuan Zhou[13], Dario Livio Longo[16], Feriel Ramdhane[16], Zhongliang Zu[17], Malvika Viswanathan[17], Pedram Parnianpour[18], Erik P. Pioro[18], Mara Quach[19,20], Martin Uecker[1], Moritz Zaiss[4]

[1]Institute of Biomedical Imaging, Graz University of Technology, Graz, Austria
[2]Physikalisch-Technische Bundesanstalt (PTB), Braunschweig and Berlin, Germany
[3]NVision Quantum Technologies, Ulm, Germany
[4]Institute of Neuroradiology, Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), University Hospital Erlangen, Erlangen, Germany
[5]School of Biomedical Engineering, Tel Aviv University, Tel Aviv, Israel
[6]Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel
[7]Lee Kong Chian Faculty of Engineering and Science, Universiti Tunku Abdul Rahman, Kajang, Malaysia
[8]Philips Singapore, Singapore, Singapore
[9]Department of Medical Physics, Memorial Sloan Kettering Cancer Center, New York, USA
[10]Department of Diagnostic Radiology, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Hong Kong
[11]Department of Imaging and Interventional Radiology, The Chinese University of Hong Kong, Hong Kong
[12]F.M. Kirby Research Center for Functional Brain Imaging, Kennedy Krieger Institute, Baltimore, USA
[13]Department of Radiology and Radiological Science, Johns Hopkins University School of Medicine, Baltimore, USA
[14]Biomedical Imaging Research Institute, Cedars-Sinai Medical Center, Los Angeles, California, USA
[15]Department of Electrical and Computer Engineering, Johns Hopkins University, Baltimore, USA
[16]Institute of Biostructures and Bioimaging, National Research Council of Italy, Torino, Italy
[17]Institute of Imaging Science, Vanderbilt University Medical Center, Nashville, USA
[18]Djavad Mowafaghian Centre for Brain Health, University of British Columbia, Vancouver, Canada
[19]Department of Biomedical Engineering and Graeme Clark Institute, The University of Melbourne, Parkville, Australia
[20]Melbourne Brain Centre Imaging Unit, The University of Melbourne, Parkville, Australia

July 6, 2026

Abstract

Purpose: Quantitative chemical exchange saturation transfer (CEST) MRI depends on Bloch-McConnell simulations, yet it is unclear how consistent implementations are across research groups. This study quantified inter-framework variability and identified dominant sources of disagreement.

Methods: Eight benchmark cases were defined and distributed online in two rounds: four continuous-wave preparations on a two-pool creatine and a five-pool white-matter model (cases 1–4), followed by Gaussian and pulsed WASABI preparations (cases 5–8). Twenty groups submitted Z-spectra for comparison.

Results: For simple two-pool continuous-wave saturation (cases 1–2), most submissions agreed within $\sim 10^{-4}$ in Z-spectra difference, but residual spread exceeded numerical precision. By contrast, marked differences were observed in the simulations for the five-pool model (Cases 3–4) and time-dependent saturation pulses (Cases 5–7). The different implementation of the magnetization transfer (MT) pool was identified as a major contributor to these differences. Overall case 5 (a single Gaussian pulse) showed the largest variability among groups; related pulse trains (cases 6–7) displayed additional sideband differences, whereas a short two-pulse WASABI scheme (case 8) agreed more closely. The final observed differences in the simulations could lead to biases of tens of percent in the fitted pool fractions or exchange rates.
Conclusion: Although the presented study helped to find common ground, Bloch-McConnell simulators from different groups are not yet interchangeable, even for fully specified protocols. Harmonized MT modeling (full xyz versus z-only), shaped-pulse handling, and clear RF power definitions are prerequisites for reproducible quantitative CEST. The open benchmark supports ongoing validation and community standardization.

Keywords: CEST, MRI, Bloch-McConnell, magnetization transfer, quantitative MRI, inter-site comparison
Correspondence: Markus Huemer, Institute of Biomedical Imaging, Graz University of Technology, Stremayrgasse 16/III, 8010 Graz, Austria. Email: `huemer@tugraz.at`
Funding: This research was funded in whole or in part by the Austrian Science Fund (FWF) 10.55776/F100800 and 10.55776/I4870. This work was partially supported by Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) grant 372486779 (SFB1340), and grant ZA 814/15-1.

1 Introduction

Chemical Exchange Saturation Transfer (CEST) MRI is a powerful imaging technique that provides contrast sensitive to exchange processes between solute and water protons. This unique sensitivity enables novel contrasts that are increasingly being explored in clinical MRI applications. Among the most commonly studied CEST contrasts is amide proton transfer weighted (APTw) imaging, which has been used in various organs and has been shown to be particularly useful in the assessment of cancer [1], neurodegenerative diseases [2], and other pathologies like ischemic stroke [3]. In the context of cancer, CEST MRI has been used to differentiate tumor progression from radiation necrosis [4], assess treatment response [5], and detect genetic mutations such as IDH in gliomas [6]. Recently, there has been growing interest in quantitative CEST approaches (qCEST), which aim to provide quantitative insights into exchange rates and solute concentrations. These approaches include Bloch-McConnell model fitting, ratiometric methods, and CEST MR fingerprinting. The analysis techniquesare based the Bloch-McConnell equations, which extend the classical Bloch equations to systems of coupled proton pools with chemical exchange [7]. For a system of N pools, the magnetization vector $\mathbf{M}$ contains the transverse and longitudinal components of each pool. During RF irradiation at offset $\Delta\omega$, the time evolution is described by the coupled linear system

$$\dot{\mathbf{M}} = \mathbf{A}(\Delta\omega, B_1, \ldots)\mathbf{M} + \mathbf{c}, \tag{1}$$

where $\mathbf{A}$ encodes relaxation, chemical exchange, off-resonance effects, and RF saturation, and $\mathbf{c}$ accounts for equilibrium magnetization. By simulating this evolution for a defined saturation scheme and a range of offsets, one obtains model Z-spectra and derived contrasts such as magnetization transfer ratio asymmetry (MTR_{asym}). Because qCEST fitting and fingerprinting pipelines depend directly on these forward simulations, differences between numerical implementations can translate into biases in estimated pool sizes and exchange rates.

Numerous simulation frameworks have been developed to perform Bloch-McConnell simulations, ranging from tools used exclusively within individual research groups to open-source platforms such as Pulseq-CEST [8], BMCTool [9], and CEST GUI [10]. Despite the availability of these tools, no systematic comparison between different simulation frameworks has been conducted. This lack of comparison is concerning because the accuracy and consistency of Bloch-McConnell simulations directly affect the results of qCEST analyses. The variability in simulation results can propagate to differences in quantitative CEST parameters. Without a ground truth for validation, it is challenging to verify the accuracy of simulation outputs, raising the possibility that discrepancies between frameworks might contribute significantly to the observed variability in qCEST-derived metrics reported in the literature.

The objective of this work is to address this gap by providing a platform to compare and ultimately validate Bloch-McConnell simulations from various CEST research groups. To achieve this, we defined two rounds of Bloch-McConnell simulation scenarios, each comprising four cases with well-defined saturation schemes and pool models. These cases were designed to capture a range of commonly encountered CEST saturation settings, including simple continuous-wave (cw) and more complex pulse trains of shaped pulses commonly used at clinical MRI scanners, as well as different pool configurations that mimic the situation from simple phantoms to brain tissue. Using this platform, we compared simulation results from up to 20 different groups, offering new insights into the consistency and reliability of Bloch-McConnell simulations across the field.

2 Methods

All necessary simulation settings and information were provided on our public GitHub repository at `https://github.com/pulseq-cest/BMsim_challenge`.

2.1 General Simulation Settings

- Fully relaxed initial longitudinal magnetization ($Z_i = 1$) before simulating each offset. Equivalently, the recovery interval between offsets must satisfy $T_{\mathrm{rec}} > 10 \cdot T_1$ of the slowest pool (e.g. water with $T_1 = 3$ s in the 2-pool model requires $T_{\mathrm{rec}} > 30$ s). All pool-model YAML files specify `reset_init_mag: True` for this purpose. The shorter recovery delays $T_{\mathrm{rec}} = 3.5$ s (cases 1–3, 5–7) or 3 s (case 4) in the reference Pulseq files are *not* part of the simulation specification; they reflect typical scanner timing between acquisitions and must not be used as a substitute for resetting $Z_i = 1$.
- Relative concentration of water is set to one ($f_{water} = 1$)
- After each saturation a delay of 6.5 ms was simulated to account for the spoiler gradient, which would normally be applied in measurements (post-preparation delay $t_{delay} = 6.5$ms).
- Gyromagnetic ratio: 42.5764 MHz/T which leads to a Larmor frequency (for 3T) of 127.7292 MHz
- Normalization scan at -300 ppm using the same preparation

2.2 Pool Models

Two different pool models were defined and used in the eight simulation cases. The first pool model is a 2-pool model and consists of a water and one single guanidine-like CEST pool. It is supposed to mimic the situation in a creatine phantom. The second pool model is a 5-pool model consisting of a water pool, an ssMT pool, an amide-like CEST pool, a guanidine-like CEST pool and a rNOE pool, in order to mimic more complex in vivo like environments. The pool model configuration files were defined in human- and machine readable yaml file format and can be found on the public GitHub repository as well as in the appendix of this manuscript.

2.3 Simulation Cases

In total, eight cases of increasing complexity were compared in two rounds on the pool models described in Section 2.2. An overview with the parameters for all cases is listed in Table 1 and the saturation pulses are shown in Figure 1.

2.3.1 Round 1

In the first round of the challenge, four cases with continuous-wave saturation and different saturation times were tested. The aim was to begin with cases that should be comparatively straightforward to reproduce and then add complexity from shorter saturation times, additional exchanging pools, and MT modeling.

- **Case 1:** A 15 s CW saturation on the 2-pool model with $B_1 = 2\,\mu T$. This case tests the steady-state for a simple creatine-like phantom and was intended as a baseline test for exchange, relaxation, offset handling, and sign conventions.
- **Case 2:** A 2 s CW saturation on the same 2-pool model with $B_1 = 2\,\mu T$. Compared with Case 1, this case emphasizes transient saturation behavior while keeping the pool model unchanged, thereby testing whether implementations agree away from the full steady-state.
- **Case 3:** The same 2 s CW saturation as in Case 2, but applied to the 5-pool white-matter model. This case tests the handling of multiple exchanging pools and, in particular, the implementation of the semisolid MT pool.
- **Case 4:** A short block-pulse preparation as used in the WASABI method [11], consisting of a 5 ms pulse with $B_1 = 3.7\,\mu T$ applied to the 5-pool white-matter model. This case probes short-pulse and off-resonance behavior in a multi-pool system, including the influence of MT and phase evolution during a brief RF event.

2.3.2 Round 2

In the second round, pulsed saturation using Gaussian and block-pulse preparations was compared on the same models. These cases were designed to test whether groups implemented shaped RF pulses, interpulse delays, and pulse trains consistently when the saturation preparation is provided in a machine-readable format.

- **Case 5:** A single Gaussian pulse with $t_p = 50$ ms and $B_{1,\mathrm{rms}} = 1.9962\,\mu T$ on the 2-pool model. The pulse shape was provided as a csv file with time and amplitude values. This isolated pulse tests interpolation, pulse handling and RF amplitude scaling, without additional complications from a pulse train.
- **Case 6:** The Gaussian pulse from Case 5 repeated 36 times with an interpulse delay $t_d = 5$ ms, for a combined T_{sat} of 1.975 s, again on the 2-pool model. This case extends Case 5 to a clinically relevant pulse train and tests accumulation of saturation and relaxation over repeated RF events.
- **Case 7:** The same Gaussian pulse train as in Case 6, but applied to the 5-pool white-matter model. This case combines shaped-pulse train implementation with the multi-pool and MT-modeling complexity introduced in Case 3.
- **Case 8:** The WASABI pulse from Case 4 ($B_1 = 3.7\,\mu T$) played out twice with an interpulse delay $t_d = 100\,\mu s$ on the 5-pool white-matter model. This short two-pulse preparation tests whether implementations handle phase accumulation and relaxation during very short delays consistently.

Table 1: Overview over the different simulation cases. CW block pulses (cases 1–4) use constant B_1 amplitude; WASABI block pulses (cases 4 and 8) use $B_{1,\mathrm{peak}}$; Gaussian pulses (cases 5–7) use $B_{1,\mathrm{rms}}$.

Case #	**pool model**	**# of pulses**	**pulse shape**	**pulse duration**	**interpulse delay**	B_1	**offsets**
1	2 pool	1	block	15 s	-	2 µT	-15:0.1:15 ppm
2	2 pool	1	block	2 s	-	2 µT	-15:0.1:15 ppm
3	5 pool	1	block	2 s	-	2 µT	-15:0.1:15 ppm
4	5 pool	1	block	5 ms	-	3.7 µT	-2:0.05:2 ppm
5	2 pool	1	Gaussian	50 ms	-	1.996 µT (rms)	-2:0.02:2 ppm
6	2 pool	36	Gaussian	50 ms	5 ms	1.996 µT (rms)	-15:0.1:15 ppm
7	5 pool	36	Gaussian	50 ms	5 ms	1.996 µT (rms)	-15:0.1:15 ppm
8	5 pool	2	block	5 ms	100 µs	3.7 µT	-2:0.05:2 ppm

2.4 Entries and participating groups

In total, 20 groups or solver teams participated in the study. Some groups submitted multiple solver variants or additional z-only MT simulations, so the number of spreadsheet result columns is larger than the number of participating groups. Additionally, some groups did not submit results for all cases, leading to the following number of submitted entries per case:

- **Case 1:** 22 entries
- **Case 2:** 23 entries
- **Case 3:** 24 entries
- **Case 4:** 23 entries
- **Case 5:** 18 entries
- **Case 6:** 19 entries
- **Case 7:** 19 entries
- **Case 8:** 19 entries

Additional information on the participating groups, solver variants, and implementation details is provided in Table 1 and the supplementary material (Additional description of simulations).

The main comparison figures exclude entries labeled as z-only MT simulations, except for the dedicated xyzMT versus zMT comparison in Section 3.2. All zMT only entries reported using a Lorentzian lineshape for the MT pool.

2.5 xyzMT vs zMT

In Bloch–McConnell simulations of magnetization transfer (MT), the semisolid pool is often modeled using only longitudinal (M_Z) magnetization, assuming that transverse components decay instantaneously due to extremely short T_2 values. This yields a simplified scalar description in which RF irradiation affects the pool through saturation governed by an effective absorption lineshape and is also known as the Henkelmann model [18]. In contrast, an xyz formulation

Table 2: Overview of participating groups, solver variants, and implementation details. Submission numbers are used as labels in the comparison figures and were assigned in the order of submission.

Subm.	Corresponding Participant	Prog. Language	Number of Pools	MT	Solver	Paper & Code
S01	**Moritz Zaiss - FAU**	Matlab 2017b / C++	dynamic	x,y,z	expm with Pade (6 iterations)	pulseq-cest matlab
S02	**Patrick Schuenke - PTB**	python	dynamic	x,y,z	expm with eig	BMCTool - `github.com/schuenke/BMCTool`
S03	**Markus Huemer - IBI TUG**	Matlab	5 fixed	x,y,z	expm with Operator Splitting	extension of [12]
S04	**Markus Huemer - IBI TUG**	Matlab	7 fixed	x,y,z	expm with eig	
S05	**Qing Zeng - KKI/JHU**	Python	dynamic	x,y,z	expm	
S06	**Nikita Vladimirov - TAU**	Python/C++	dynamic	x,y,z	Pade	open-py-cest-mrf [13] `github.com/momentum-laboratory/molecular-mrf`
S07	**Zhongliang Zu - VUMC**	Matlab	dynamic	only z	expm with eig	
S08	**Kexin Wang - KKI/JHU**	Matlab	fixed	x,y,z	expm	`github.com/Kexin-Wang/BMsim`[14]
S09	**Dario Longo - CNR**	Matlab 2021b	fixed	x,y,z	expm	extension of `github.com/cest-sources/BM_sim_fit`[15]
S10	**Moritz Fabian - FAU**	Matlab 2019b/2022b	dynamic	x,y,z	RKDP	`gitlab.com/mrilab/sim/bmsim.git`
S11	**Beomgu Kang - KKI/JHU**	Matlab / python	dynamic	x,y,z	expm	`github.com/Heo-Group`[16]
S12	**Huabin Zhang - HKU**	Matlab 2023a	dynamic	x,y,z	expm	`github.com/JianpanHuang/BMsim_challenge`
S13	**Jian Hou - CUHK**	Matlab 2023a	dynamic	only z	expm	
S14	**Viktoria Buchegger - IBI TUG**	C	fixed	x,y,z	RKDP	BART [17] `codeberg.org/mrirecon/bart`
S15	**Mara Quach - UniMelb**	Julia v1.11	dynamic	x,y,z	Rosenbrock	`github.com/maraquach/CESTSim.jl`
S16	**Felix F. Zimmermann - PTB**	python (torch)	dynamic	x,y,z	expm	`github.com/fzimmermann89/mr2`
S17	**Ouri Cohen - MSK**	pytorch	dynamic	x,y,z	expm	
S18	**Jingpu Wu - JHU**	python	dynamic	x,y,z	expm	`github.com/JimmyWu990110/BMsim_challenge_Jingpu`
S19	**Pedram Parnianpour - UBC**	python	dynamic	x,y,z	expm	`github.com/Pedram-Parnianpour/bmsim-challenge-solver.git`
S20	**Swee Qi Pan - UTAR**	Matlab	dynamic	x,y,z	expm	`github.com/sweeqipan/BMsim_SQPan_YKTee`

explicitly includes transverse components, enabling the modeling of coherent effects such as off-resonance excitation and phase evolution. The z-only approach is computationally efficient and frequently used in simulations for MT and also CEST experiments. In the original formulation of this study, the way to simulate the MT pool was not explicitly stated, but results of the first round showed significant differences between xyzMT and zMT models, for the same parameters. An example of this is shown in Section 3.2 of the results. These differences prompted a switch in the study to explicitly require xyzMT simulations. The zMT simulations are kept in the data source, but are not shown for all results.

2.6 Data analysis

Magnetization transfer ratio asymmetry (MTR_{asym}) was computed from each Z-spectrum as $Z(-\Delta\omega) - Z(+\Delta\omega)$ for positive offsets $\Delta\omega$. Inter-submission spread was quantified point-wise at each offset as the absolute difference between the maximum and minimum value across included submissions, separately for the spectra and MTR_{asym}; for each case we report the maximum and median spread across offsets. The normalization measurement at -300 ppm was excluded from Z-spectrum spread calculations. In the main comparison figures, spectra and asymmetry curves are also shown as differences to submission S02 (BMCTool), which was used as a documented Pulseq-based reference implementation. It generally lies close to the main cluster of submissions, although it is not the median submission for every case.

2.7 Numerical implementation checks

Two additional analyses were performed to illustrate how numerical settings can affect pulsed CEST simulations. For the BART submission (S14), single- and double-precision variants were compared with the reference S02 for Cases 5 and 6 (Figure 8). For the operator-splitting MATLAB submission S03, Case 6 Z-spectra were calculated using multiple fixed timesteps from 500 to 5 μs (Figure 9).

3 Results

The results of the study were collected and are available online at: `https://docs.google.com/spreadsheets/d/1JN7VN-f1ktDrJgokbOFlUFwkHOMWYlPA_jSfnQoFOVc/`. An evaluation script is available at `https://colab.research.google.com/drive/1csiIjK-fiftdb7OwvJ84gWuv8lLADgv7`.

Unless noted otherwise, the main comparison figures follow the analysis described in Section 2.6 and exclude entries labeled as z-only MT simulations. These entries are retained in the data source and shown separately for the MT comparison.

3.1 Round 1

All results for Cases 1 and 2 are shown in Figure 3. Case 1 showed the closest agreement among the Round 1 scenarios. After excluding the normalization offset, the maximum spread across the submitted Z-spectra was 2.7×10^{-3}, while the median point-wise Z spread was 5.4×10^{-5}. The corresponding maximum and median MTR_{asym} spreads were 8.4×10^{-5} and 9.4×10^{-6}, respectively.

For Case 2, the shorter saturation time increased the sensitivity to implementation details. Most submissions followed the same overall Z-spectrum shape, but one or more traces deviated visibly over parts of the offset range, resulting in a maximum point-wise Z spread of 0.43 and a median Z spread of 4.0×10^{-3}. The maximum and median MTR_{asym} spreads were 0.0095 and 3.8×10^{-5}, respectively, indicating that the largest differences were localized and not representative of all offsets.

All results for Cases 3 and 4 are shown in Figure 5. Introducing the 5-pool white-matter model in Case 3 led to broader but more consistent deviations around the offsets affected by the CEST and MT pools. Excluding z-only MT entries, the maximum and median Z spreads were 0.031 and 0.027, and the maximum and median MTR_{asym} spreads were 6.5×10^{-4} and 2.7×10^{-4}, respectively. Case 4, which used a short WASABI-like block pulse on the same pool model, showed visible differences in the Z-spectra, with maximum and median Z spreads of 0.018 and 6.9×10^{-3}, and maximum and median MTR_{asym} spreads of 0.0016 and 7.3×10^{-4}, respectively.

3.2 MT: XYZ vs Z

Selected entries are shown for Case 3 to highlight the difference between full xyz MT simulations and z-only MT simulations in Figure 4. The z-only MT simulations formed a distinct response compared with the full xyz MT implementations, especially in spectral regions where the semisolid pool contributes strongly. Previous analysis [19] showed that the difference betweet z-only and xyz MT simulations can lead to significant differences in fitted pool fractions and exchange rates, which can be as high as 30% for the semisolid pool fraction or 50% for the exchange rate.

Because this difference reflects a modeling choice rather than only a numerical solver difference, z-only MT entries were excluded from the main comparison figures.

3.3 Round 2

All results for Cases 5 and 6 are shown in Figure 6. Case 5, the single Gaussian pulse, produced the largest absolute Z-spectrum spread of all cases. Although many submissions agreed closely near the central part of the spectrum, one or more submissions showed pronounced deviations, leading to a maximum Z spread of 1.45 and a median Z spread of 6.3×10^{-3}. The MTR_{asym} spread was much smaller, with maximum and median values of 0.0026 and 9.1×10^{-4}, respectively, because the largest spectrum deviations were largely symmetric.

In Case 6, where the same Gaussian pulse was repeated as a pulse train, the deviations persisted but changed in spectral distribution. The maximum and median Z spreads were 0.76 and 7.6×10^{-3}, and the maximum and median MTR_{asym} spreads were 0.0089 and 6.8×10^{-5}, respectively. Differences were especially visible away from the central offsets, consistent with sensitivity to pulse-shape scaling, interpolation, and repeated application of relaxation and exchange during the interpulse delays.

All results for Cases 7 and 8 are shown in Figure 7. Case 7 combined the Gaussian pulse train with the 5-pool model and showed maximum and median Z spreads of 0.031 and 7.2×10^{-3}, and maximum and median MTR_{asym} spreads of 9.0×10^{-4} and 6.9×10^{-4}, respectively. Compared with Case 6, the multi-pool model shows much closer agreement.

Case 8 showed a different pattern. The short two-pulse WASABI-like preparation produced visible Z-spectrum differences, with maximum and median Z spreads of 0.093 and 0.048. The MTR_{asym} curves agreed more closely, with maximum and median spreads of approximately 10^{-3} and 4.7×10^{-4}, respectively.

3.4 Numerical precision and time-step convergence

The BART comparison in Figure 8 shows that single- and double-precision arithmetic can produce visibly different spectra for shaped RF preparations, even when the underlying solver and pool model are identical. For the single Gaussian pulse (Case 5), the maximum point-wise difference to S02 was 0.0033 for single-precision BART and 1.5×10^{-4} for the double-precision variant. The effect was much stronger in the pulse train (Case 6), where single-precision BART deviated from S02 by up to 0.24 in Z, whereas double-precision BART remained within approximately 1.4×10^{-3}.

Figure 9 shows the corresponding fixed-timestep analysis for submission S03 in Case 6. A timestep of 500 μs produced a clearly different Z-spectrum, reducing the timestep to 100 μs produces a normal looking spectrum, which still deviates from the finest timestep of 5 μ by over 10%. Halfing the timestep to 50 μs produces a spectrum reduces the deviation to less than 5% and further reducing the timestep to 10 μs produces a spectrum that is indistinguishable (at the shown scale) from the finest timestep.

4 Discussion

In this study we showed that a main cluster of agreeing simulations could be found consisting of 16 groups in the cw case, and 10 groups in the pulsed CEST case, which tentatively allows us to formulate a consensus on simulations. However, this study also shows that Bloch-McConnell simulations are not automatically interchangeable across independent implementations, even when the nominal pool model, RF preparation pulse trains, offsets, and initialization conditions are specified. The degree of agreement depended strongly on the simulation case. The long CW 2-pool case showed close agreement, whereas shorter saturation, shaped pulses, pulse trains, and the 5-pool white-matter model all increased the observed variability. These findings indicate that benchmark cases need to include both simple reference scenarios and more clinically realistic preparations; agreement in a steady-state-like 2-pool case alone is not sufficient to validate a simulation framework for quantitative CEST applications.

The first round separated several sources of variability. Case 1 was intended as the simplest baseline and showed the smallest spread, suggesting that most implementations handle the basic Bloch-McConnell evolution, offset convention, and relaxation/exchange terms similarly under near steady-state conditions. The larger spread in Case 2 suggests that transient behavior is more sensitive to numerical and implementation choices. Since the pool model and RF amplitude were unchanged between Cases 1 and 2, these differences point to details such as matrix exponentiation, ODE integration tolerances, pulse/discretization conventions for CW irradiation, or the precise handling of delays. The lower spread in Case 1 may reflect that the long saturation approaches a common steady-state solution, whereas the shorter preparation in Case 2 preserves differences in transient evolution.

The 5-pool model introduced a second major source of variability. The comparison of full xyz MT and z-only MT implementations showed that these approaches produce systematically different spectra and should not be mixed without explicit labeling. This is expected because a z-only semisolid pool model approximates the MT pool using longitudinal saturation and an absorption lineshape, whereas a full xyz implementation evolves transverse components explicitly.

Both approaches may be useful in specific contexts, but they are not equivalent for the benchmark cases used here. For this reason, z-only MT results were retained in the data source but excluded from the main figures. The lineshape of the ssMT effect is an open question in the field, with Lorentzian, super Lorentzian, Gaussian, and other lineshapes being used in different studies [20]. Depending on the investigated tissue and the specific application, different lineshapes may be more appropriate. The present benchmark does not address this question, but it highlights that the choice of MT modeling can have a substantial impact on the simulated spectra and should be clearly specified in future studies. Furthermore, these lineshapes are often implemented using numerical approximations, such as discretization of the lineshape or interpolation around pole locations. This can introduce additional variability, and possibly errors if not implemented carefully.

The second round showed that shaped RF preparations are another important source of disagreement. The single Gaussian pulse produced the largest absolute Z-spectrum spread, even though the pulse shape was provided in a machine-readable file. This suggests that providing the shape alone does not fully eliminate ambiguity: interpolation of the RF waveform, time discretization, conversion between peak and rms B_1, treatment of phase, and the specific pulse integration can all affect the result. The pulse-train cases further amplify these choices because small differences are repeated many times and can accumulate across interpulse delays. The multi-pool model in Case 7 shows much closer agreement, indicating that the differences are not caused by the multi-pool model itself, but by the shaped pulse implementation and the long T1 and T2 values of water for the 2 pool model. The long relaxation times of water in the 2 pool model lead to a slow decay of the magnetization, which necessitates accurate simulation throughout the pulse train, as any small difference in the magnetization evolution can accumulate and lead to significant differences in the final spectrum.

4.1 Solver choices and differences

In general, the results show that the same solution can be reached with different numerical strategies, provided that the implementations are sufficiently accurate for the RF preparation and pool model. Closely agreeing submissions used different approaches, including matrix exponentiation, Padé approximations, eigenvalue decompositions, ODE solvers, and operator splitting. Thus, the solver class itself does not appear to determine the result uniquely. Instead, deviations are more likely caused by the details of how the solver is applied, such as time-step selection, waveform discretization, convergence criteria, and the handling of very short relaxation constants.

One important distinction is whether the simulation uses fixed or adaptive time steps. A fixed time-step implementation can be efficient and reproducible, but the chosen step size must be short enough to resolve the fastest dynamics in the system. This becomes particularly relevant for shaped pulses and for full xyz MT implementations, where very short T_2 values lead to rapidly decaying transverse components. If these dynamics are under-resolved, the resulting spectrum can deviate even when the pool parameters are nominally identical. Conversely, very small fixed time steps may increase memory use and computation time, especially for long pulse trains and many offsets.

Adaptive ODE solvers and matrix-exponential approaches avoid some of these limitations, but introduce their own implementation choices. ODE solvers depend on tolerances, step sizes, and event discretization. Matrix-exponential approaches depend on how the RF waveform is divided into constant-amplitude intervals and how the exponential is evaluated, for example by Padé approximation or eigenvalue decomposition. For well-conditioned problems and carefully implemented solver choices these approaches should converge to the same solution, but differences can appear for long transient solutions with slow decays, when the RF pulse is coarsely sampled, when the ODE becomes stiff due to very short decay times, or approximations are used without convergence checks.

Figure 8 provides a concrete example of this issue for the BART implementation. In Case

6, single-precision arithmetic led to large deviations from the reference, whereas the additional double-precision submission agreed much more closely with S02. This indicates that numerical precision can be a limiting factor in long pulse-train simulations, independent of the chosen solver class. Figure 9 shows a related effect in Case 6for fixed-step operator-splitting timescale in S03: an overly coarse timestep of 500 μs produced a clearly incorrect spectrum, while progressively smaller timesteps converged toward the same solution. These findings support the view that solver accuracy must be validated for the specific RF preparation and exact numerical settings are important to achieve reproducible results.

The benchmark therefore suggests that future simulation reports should describe not only the mathematical model, but also the numerical propagation strategy. In particular, time-step strategy, RF resampling, solver tolerances or matrix-exponential method, and convergence checks should be documented to allow a more detailed comparison of simulation results.

4.2 Implications for qMRI and future Bloch-McConnell simulations

The relationship between Z-spectrum spread and MTR_{asym} spread was case dependent. In some cases, large Z-spectrum differences produced comparatively small MTR_{asym} differences, consistent with approximately symmetric deviations. In other cases, especially when the 5-pool model was combined with pulsed saturation, differences became strongly asymmetric and led to larger MTR_{asym} variability. This is important for qCEST because many downstream analyses use asymmetry or fitted pool parameters rather than the raw Z-spectrum. A deviation that appears modest in Z can therefore still be relevant if it affects the spectral asymmetry near the fitted pool resonance.

The present comparison does not yet assign all deviations to a single cause. Several factors can interact, including RF definitions, MT-pool formulation, solver choice, implementation and accuracy, and waveform discretization. In addition, some differences arose from ambiguity in the original challenge description rather than from an error in a particular simulation package. The benchmark therefore serves two purposes: it identifies cases where implementations diverge, and it tested which parts of the simulation specification need to be stated for reproducible quantitative CEST.

A practical implication is that future CEST simulation studies should report implementation details that are often omitted, including whether MT pools are modeled as full xyz or z-only systems, how shaped RF pulses are discretized and scaled, and how magnetization is initialized between offsets. The challenge repository has already been updated to clarify several of these points. Continued use of shared machine-readable definitions, together with reference results and open analysis scripts, should make it possible to distinguish model choices from unintended implementation differences.

As a reference simulation we suggest the BMCTool results (S02), which showed no significant deviation from the main cluster of results. The tool is easy to use with its Python interface and support for pulseq files. Therefore, comparing against BMCTool provides a quick and easy check for reasonable results, with small deviations being possible due to the above mentioned implementation details.

5 Conclusion

This benchmark shows that independent Bloch-McConnell simulation frameworks can agree closely for simple reference cases, but may diverge substantially for transient saturation, shaped RF pulses, pulse trains, and multi-pool models with MT. Clear specification of MT modeling, RF power definitions, pulse discretization, and initialization conditions is therefore essential for reproducible quantitative CEST. For newly implemented simulations we suggest comparing against the BMCTool results (S02). The provided open challenge data and analysis scripts provide a basis for continued community validation and for developing more standardized CEST simulation practice.

Data Availability Statement

The benchmark specifications, pool-model definitions, and sequence files are available in the public GitHub repository at `https://github.com/pulseq-cest/BMsim_challenge`. The Python analysis scripts used to parse the results, compute summary metrics, and generate the manuscript figures are available in the same repository. During the study, submitted Z-spectra were collected in a shared Google spreadsheet:
(`https://docs.google.com/spreadsheets/d/1JN7VN-f1ktDrJgokbOFlUFwkHOMWYlPA_jSfnQoFOVc/`).
The final curated dataset, including the submitted simulation results and derived analysis tables used in this manuscript, will be archived on Zenodo before publication.

Author contributions

M.Z. and P.S. designed the study. M.Z., and P.S. defined the simulation cases and specifications. M.Z. and P.S. coordinated the study and collected results. M.H. performed the analysis and wrote the initial paper draft. All named authors submitted their results and contributed to writing, editing and reviewing of the manuscript.

Conflict of interest

The authors declare no potential conflict of interests.

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A Description of the pool models

Table 3: Parameters for the 2 pool model

Pool	**f**(1)	$\mathbf{T_1}$(s)	$\mathbf{T_2}$ (s)	**k** (Hz)	**dw** (ppm)
water	1	3	2	-	-
creatine	5e-4	1.05	0.1	50	1.9

Table 4: Parameters for the 5 pool model

Pool	**f**(1)	$\mathbf{T_1}$(s)	$\mathbf{T_2}$(s)	**k** (Hz)	**dw** (ppm)
water	1	1	0.04	-	-
MT	0.1351	1	4e-5	30	-3
amide	9.009e-4	1	0.1	50	3.5
guanidine	9.009e-4	1	0.1	1000	2
NOE	4.5e-3	1.3	0.005	20	-3

B Additional description of simulations

S01 – M. Zaiss – Pulseq-CEST

The submission used the MATLAB implementation of Pulseq-CEST [8], which combines a Pulseq-based sequence parser with a C++ Bloch–McConnell simulation core. Pool parameters were read from the YAML model files provided with the challenge, and saturation schemes were simulated from the supplied Pulseq `.seq` files. Magnetization evolution was computed sequentially for each sequence block (RF, delays, and gradients). For piecewise-constant time intervals, the affine Bloch–McConnell system was propagated using a Padé approximation to the matrix exponential. Pools were assembled dynamically from the YAML specification, including separate water, CEST, NOE, and full xyz MT compartments. Shaped RF pulses were resampled according to the maximum number of samples defined in the model file. The Pulseq-CEST code and documentation are available at `https://pulseq-cest.github.io/`.

S02 – P. Schuenke – BMCTool

BMCTool is a purely Python-based Bloch–McConnell simulation framework for CEST and related multi-pool exchange experiments. Simulations are driven by Pulseq `.seq` files (parsed via PyPulseq), which provide the full event timing (RF, gradients, delays, ADCs), while scanner settings and pool parameters are specified in YAML configuration files. The magnetization evolution is computed block-by-block: RF events are optionally resampled to a user-defined maximum number of samples and applied piecewise; delays are simulated as free precession/relaxation; and spoiler gradients are modeled as complete transverse spoiling. For each time interval, the affine Bloch–McConnell system $\dot{\mathbf{M}} = \mathbf{AM} + \mathbf{c}$ is propagated using a Padé approximation to the matrix exponential (with the steady-state shift computed from $\mathbf{A}^{-1}\mathbf{c}$ via a pseudoinverse), enabling stable simulation of arbitrary Pulseq-defined irradiation schemes.

S03 & S04 – M.Huemer & R.Stollberger and M.Huemer & R.Stollberger 2

Simulation based on work published in [12], written in MATLAB. The second submission does not use the operator splitting proposed in [12], as a check on the influence this has on the simulation results. As described in the paper, a predefined constant time-step is used, and all offsets are solved in parallel. This can lead to memory problems and limit the accuracy with long saturation times and many offsets. The simulation and derivatives are used in [21][22] and the simulation source code is available in the data to reproduce these results `https:`

//doi.org/10.5281/zenodo.15768062.

S06 – N. Vladimirov & O. Perlman - open-py-cest-mrf
The simulator was built with a C++ backend, a Pulseq sequence specification, and a Python front end that supports parallel computation. The backend closely follows the Pulseq-cest implementation and provides an interface to facilitate MRF simulations, streamlining parameter modifications. The Bloch-McConnell equations are solved in an iterative manner using the Pade approximation for matrix exponentiation, with a SWIG-based wrapper that enables Python integration. The Python front end is tailored for MRF simulations, using BMCTool definitions to set up and modify parameters on the fly while efficiently handling C++ function calls [13].

S09 – D. L. Longo & F. Ramdhane
Simulation based on the Bloch-McConnell work published in [15], written in Matlab by adapting the code available at `https://github.com/cest-sources/BM_sim_fit`. The BM equations have been solved analytically.

S12 – H.Zhang & J.Huang - CESTsimu
CESTsimu is an open-source GUI-based CEST simulation tool, written in MATLAB. It is compatible with Pulseq-CEST, able to simulate 1D/2D CEST under B_0/B_1 inhomogeneities. The simulation solver is implemented through numerical iteration of the Bloch-McConnell equations. RF events are defined using cell arrays, where each cell represent either a continuous wave or a shaped pulse. The solver processes each RF event sequentially. The time step is specified within each pulse cell. For each time interval, the evolution is modeled in matrix form as $dM/dt = AM + c$, and the equations are solved using two built-in MATLAB functions: `mldivide()` and `expm()`. The solver code along with its demo are available at: `https://github.com/JianpanHuang/BMsim_challenge`.

S14 – V. Buchegger, M. Juschitz, N. Scholand & M. Uecker – BART
The BART submission used the Bloch–McConnell simulation implementation in the Berkeley Advanced Reconstruction Toolbox (BART). The BART toolbox is a free and open-source image–reconstruction framework for Computational Magnetic Resonance Imaging [23]. The Bloch–McConnell simulation is written in C and solves the Bloch–McConnell equations using a Runge–Kutta–Dormand–Prince scheme [17]. The main submission (S14) used single-precision floating point arithmetic; an additional variant (S14b) was provided using double precision. The source code is available at `https://codeberg.org/mrirecon/bart`.

S15 – M. Quach – CESTSim.jl
CESTSim.jl provides a Julia-based implementation for Bloch-McConnell simulations, with flexible choices of ODE solvers afforded by DifferentialEquations.jl. For the submissions to this challenge, the ODE solver of choice was a 5th-order Rosenbrock-Wanner method [24]. To demonstrate the impact of the choice of solver, an additional submission derived from using a 2/3rd-order Rosenbrock method (*Rosenbrock23()*) was provided to Case 2 (submission S15b in Figure 3). For 5-pool cases, it was assumed that all solute pools underwent exchange with water and water only (i.e., no cross-exchange between non-water pools). The relative tolerance for cases 1-4 (cw) and cases 5-8 (pulsed) were $1e-3$ and $1e-5$, respectively. The default absolute tolerance of $1e-6$ was used for all cases. For cases 5-8, the RF trains were interpolated and scaled using a constant (0th-order) B-spline. For each frequency offset, the entire RF train including pulse delays were interpolated and the system was solved across the entire saturation and spoiling delay duration, instead of block-by-block. In general, CESTSim.jl supports multi-pool simulations and the construction of complex saturation and imaging sequence protocols (and flexible interpolation options) for simulation. Other conditions such as $\delta\omega$ shift, spoiling, and

incomplete recovery can also be incorporated into the simulations.

S16 – F. Zimmermann et al. – MRpro

MRpro [25] is an open-source PyTorch-based framework for MR image reconstruction, quantitative parameter estimation, and deep learning. Its Bloch–McConnell simulator is GPU-accelerated and fully differentiable with respect to sequence and tissue parameters. It propagates the affine system $\dot{\mathbf{M}} = \mathbf{AM} + \mathbf{c}$ via matrix-exponential evolution for each sequence block, falling back to an augmented homogeneous formulation when systems are singular. The simulator supports multi-pool exchange, static off-resonance, and relative B_1 inhomogeneity through modular blocks including constant RF, piecewise-shaped RF, delays, gradients with isochromat dephasing, spoiling, and readout. For piecewise-shaped pulses, the evolution is computed sample-by-sample with chunked propagation and activation checkpointing to reduce memory use, including in the long pulse-train benchmark cases 6 and 7.

S17 – O. Cohen

The simulator was implemented in pyTorch for differentiability and efficient GPU-usage. A helper script was used to interface between the YAML configuration files and the simulator parameters. Bloch-McConnell equations are solved using the pyTorch matrix exponential (`torch.matrix_exp()`) and `torch.linalg.solve()` for the steady-state term. The simulator supports multi-pool exchange for arbitrary number of pools and flexible acquisitions using user-defined acquisition schedules.

S19 – P. Parnianpour & E. P. Pioro

The BM equations ($dM/dt = A \cdot M + b$) were solved analytically using the matrix exponential: $\mathrm{M(t)} = \mathrm{expm}(A \cdot t) \cdot (M_0 - M_{ss}) + M_{ss}$ where $M_{ss} = -A^{-1}b$ is the steady state (via `numpy.linalg.lstsq`). The matrix exponential uses `scipy.linalg.expm` (Al-Mohy & Higham 2009, scaling-and-squaring, Padé order selected automatically based on matrix norm up to m=13). No homogeneous augmentation was used. For shaped pulses (cases 5–7), the Gaussian envelope was sampled into 200 steps matching the .seq file. Case 8 uses block pulses with a 100μs interpulse delay. For pulse trains (cases 6–7: 36 pulses), the full-pulse propagator was precomputed once per offset and applied 36 times, reducing expm calls by 36×.

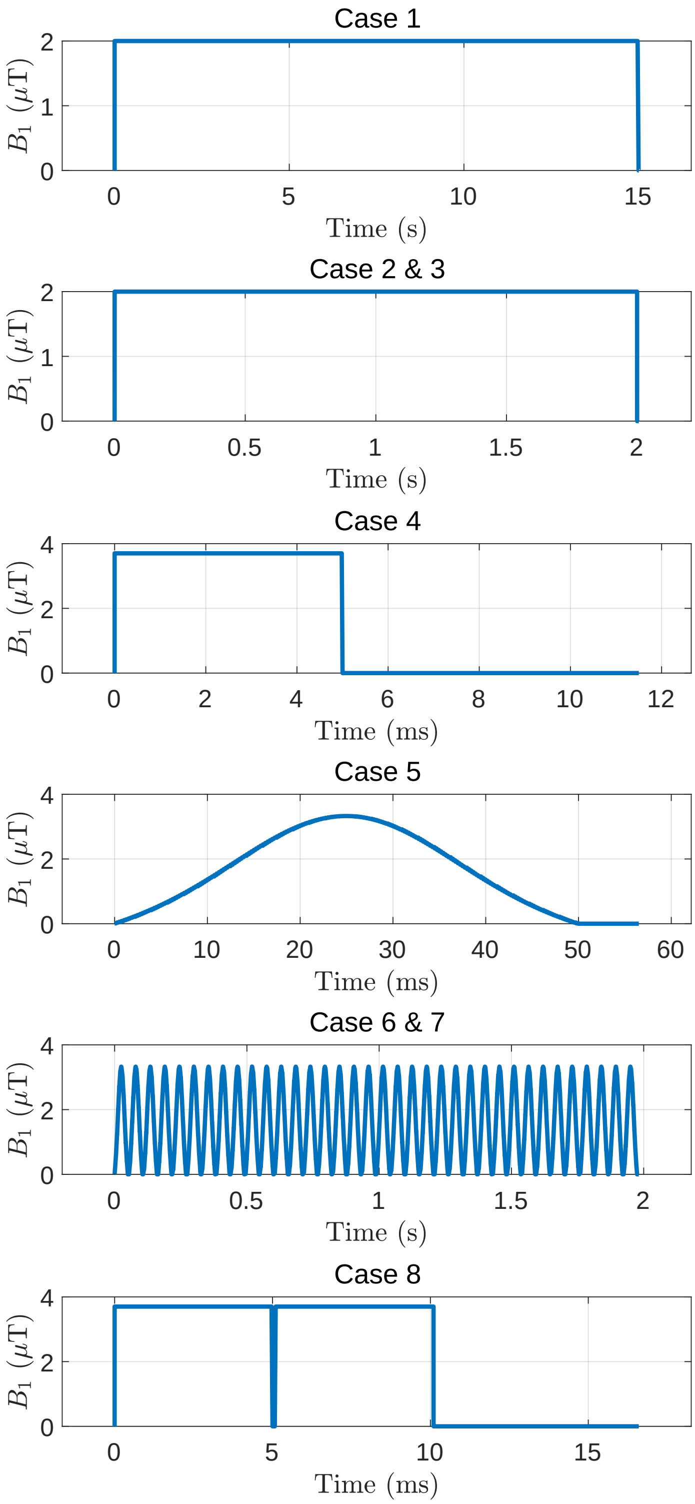

Figure 1: Saturation types for all tested cases. Cases 1 to 4 use CW saturation with different saturation times. Cases 5, 6 and 7 use one or multiple Gaussian pulses and Case 8 repeats the CW saturation of Case 4 with a short interpulse delay. Note the spoiler delay after the pulses visible for short saturation cases.

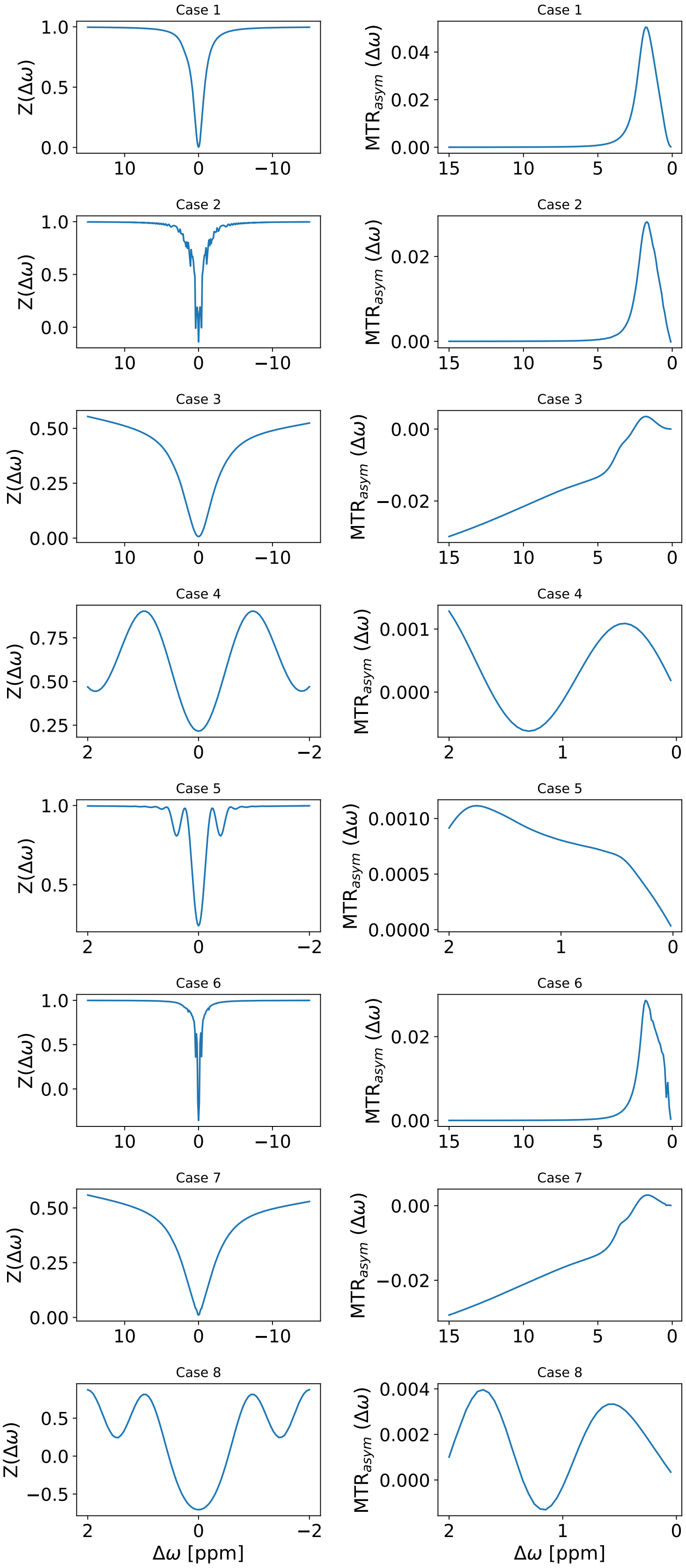

Figure 2: Exemplary z-spectra and MTR_{asym} for all eight tested cases, simulated with the MATLAB pulseq-cest implementation.

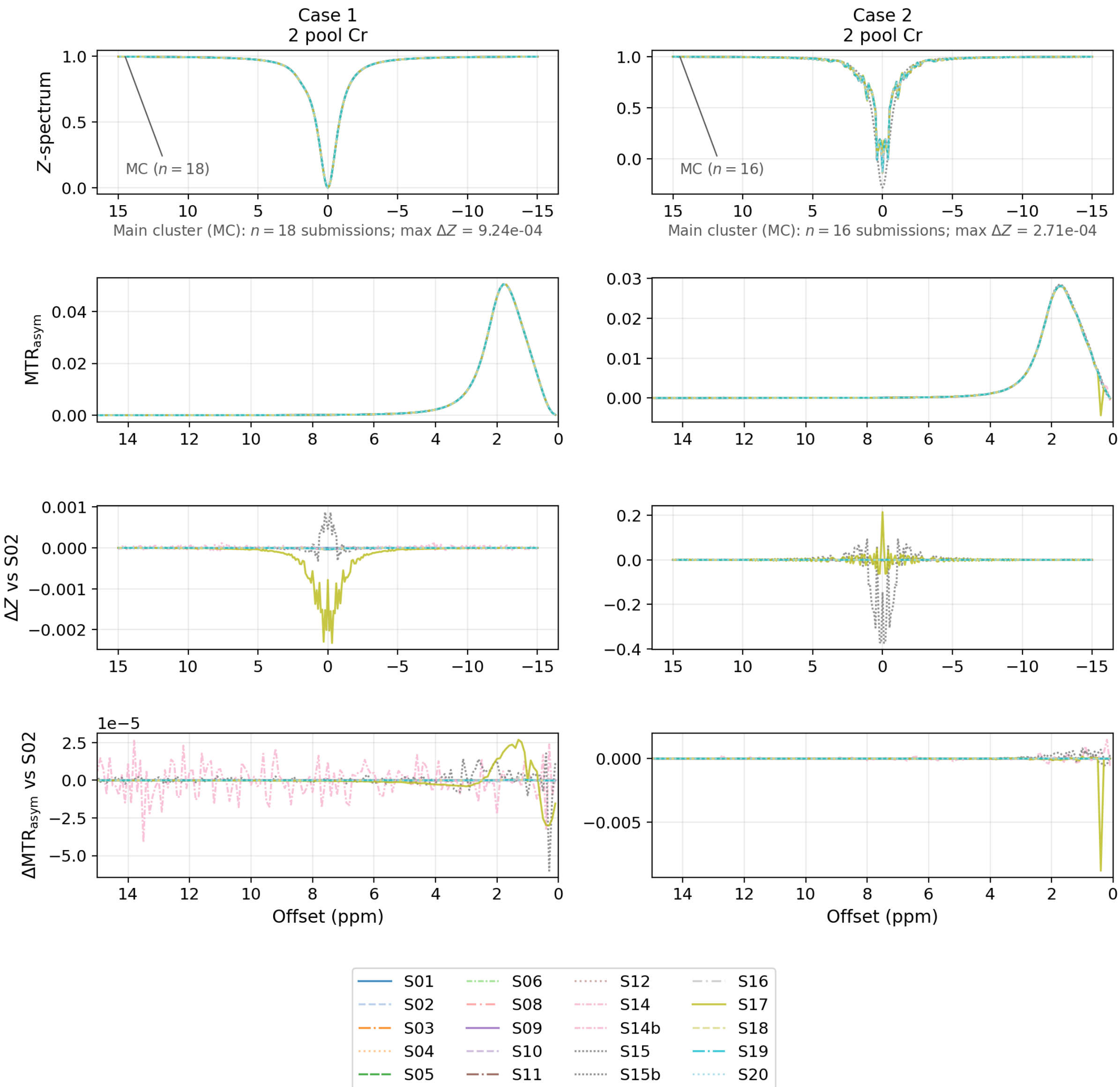

Figure 3: Cases 1 and 2: Z- and MTR_{asym} spectra with differences to S02. S15 and S17 show the largest deviations in Case 1. S14b represents a solver variant of S14, using double precision instead of single precision. In Case 2, S15b denotes an alternative solver variant of S15, which shows the largest deviation from the main cluster. More information on these solver variants can be found in the additional information.

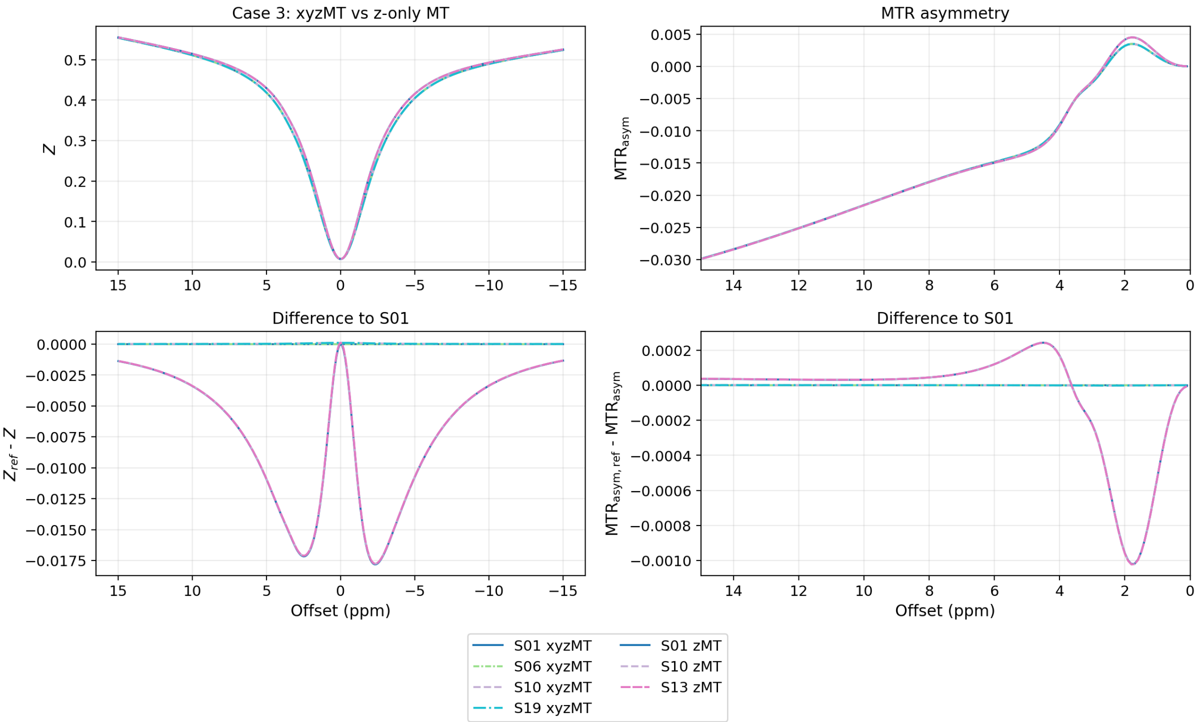

Figure 4: Selected entries for Case 3 showing full xyzMT simulations (S01, S06, S10 and S19) in comparison with z-only MT simulations (S01, S10 and S13). The xyzMT entries closely match with the chosen reference (here S01 xyzMT). The zMT entries show a significant difference to the reference, with close agreement between each other.

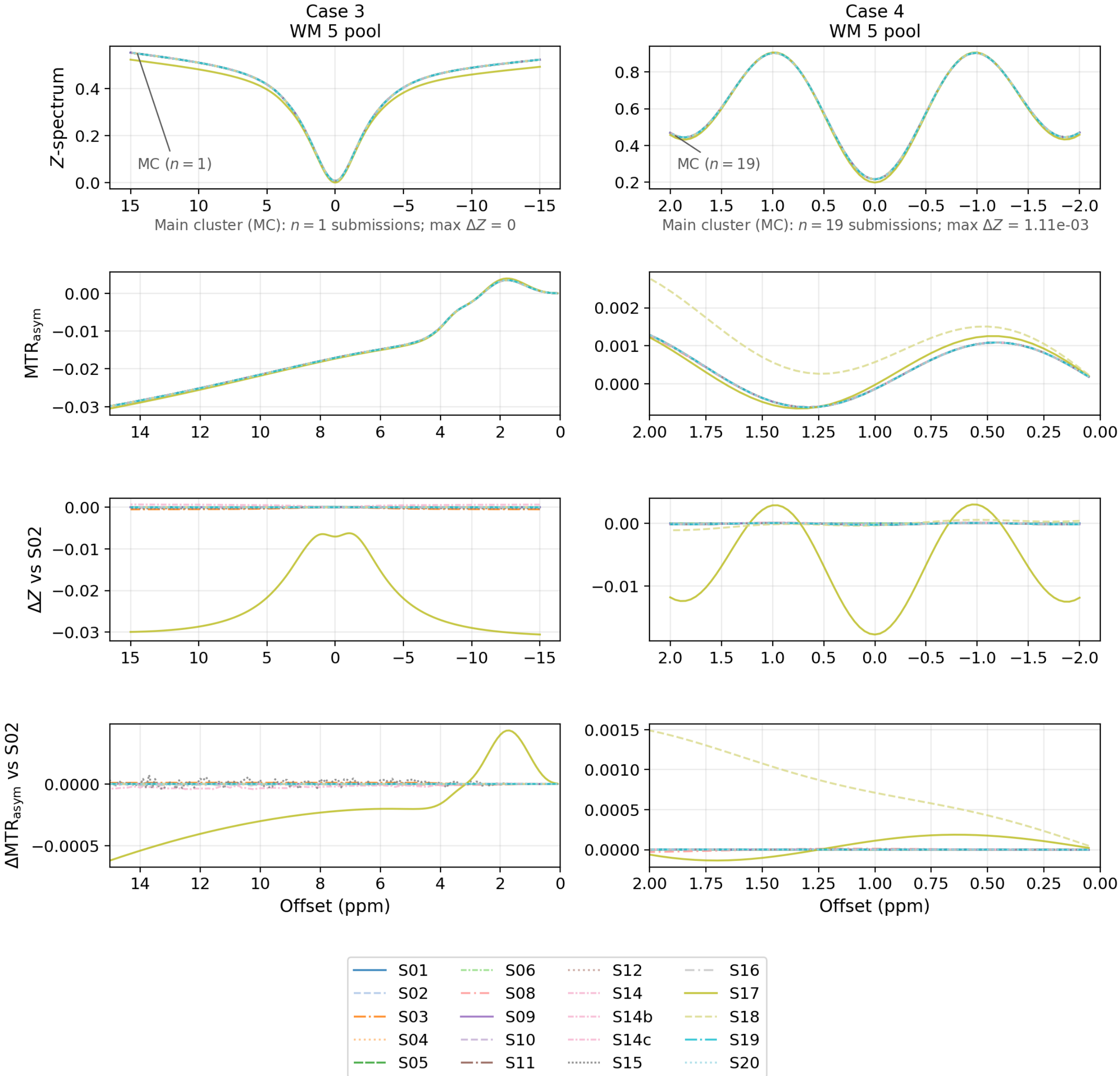

Figure 5: Cases 3 and 4: simulated Z- and MTR_{asym} spectra with differences to S02. S17 shows the largest deviations in both cases; in Case 4, S18 also differs strongly in MTR_{asym}.

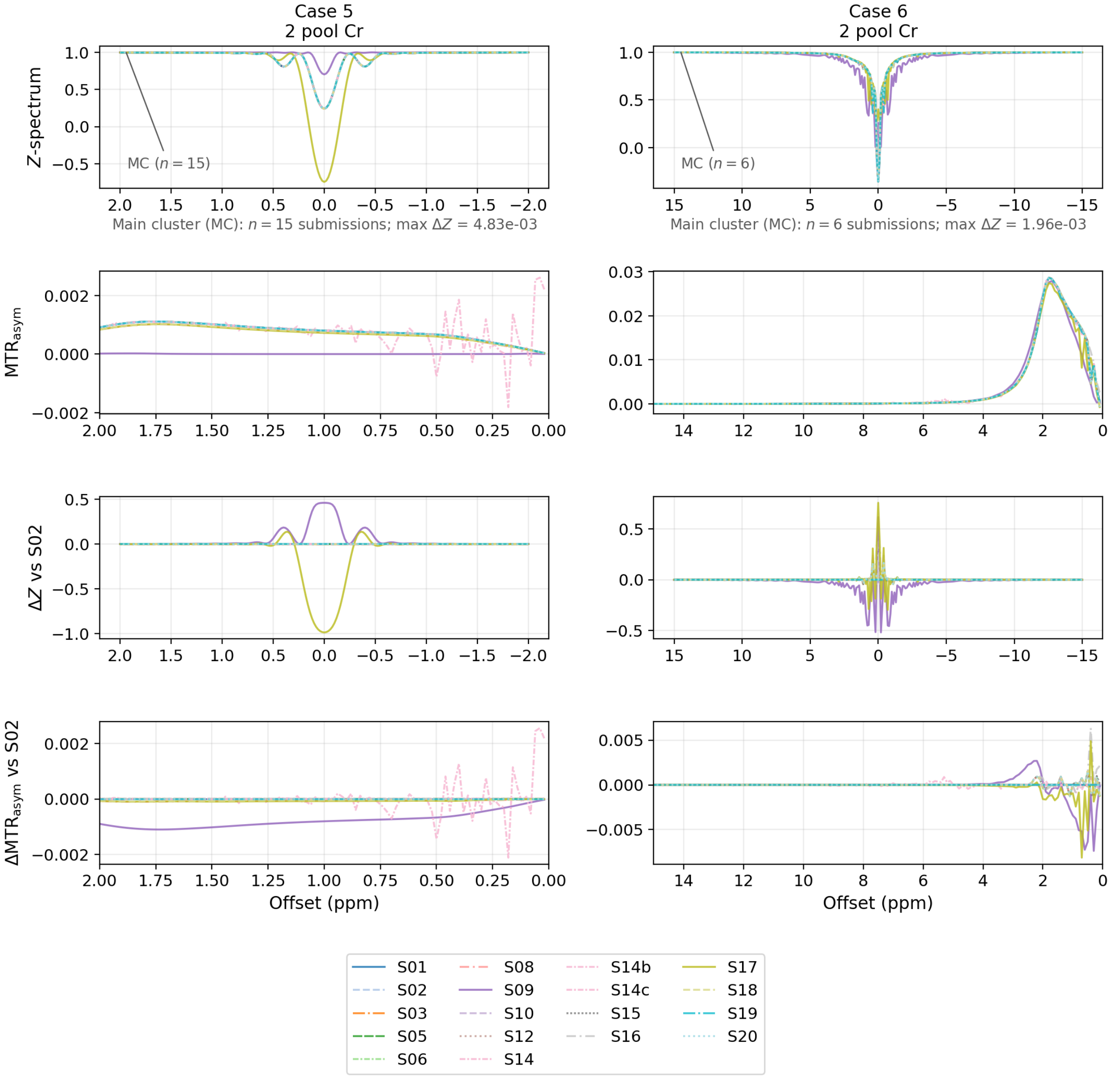

Figure 6: Cases 5 and 6: simulated Z- and MTR_{asym} spectra with differences to S02. S09 and S17 show the largest deviations in Case 5; similar separation is seen in Case 6, although the main cluster is less clearly defined.

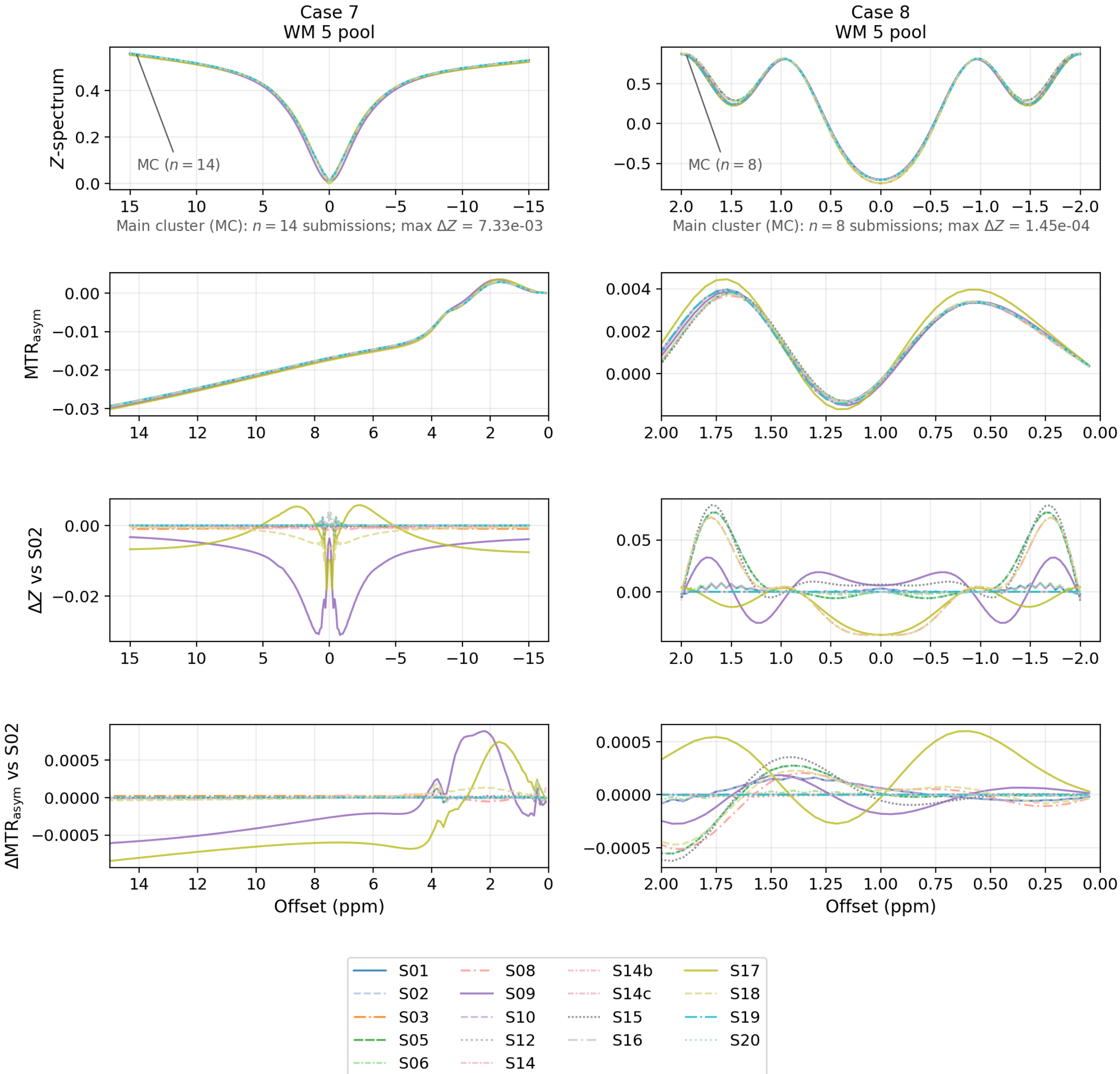

Figure 7: Cases 7 and 8: simulated Z- and MTR_{asym} spectra with differences to S02. In Case 7, S09, S18, and S19 deviate most strongly; Case 8 shows broader spread with several separated traces.

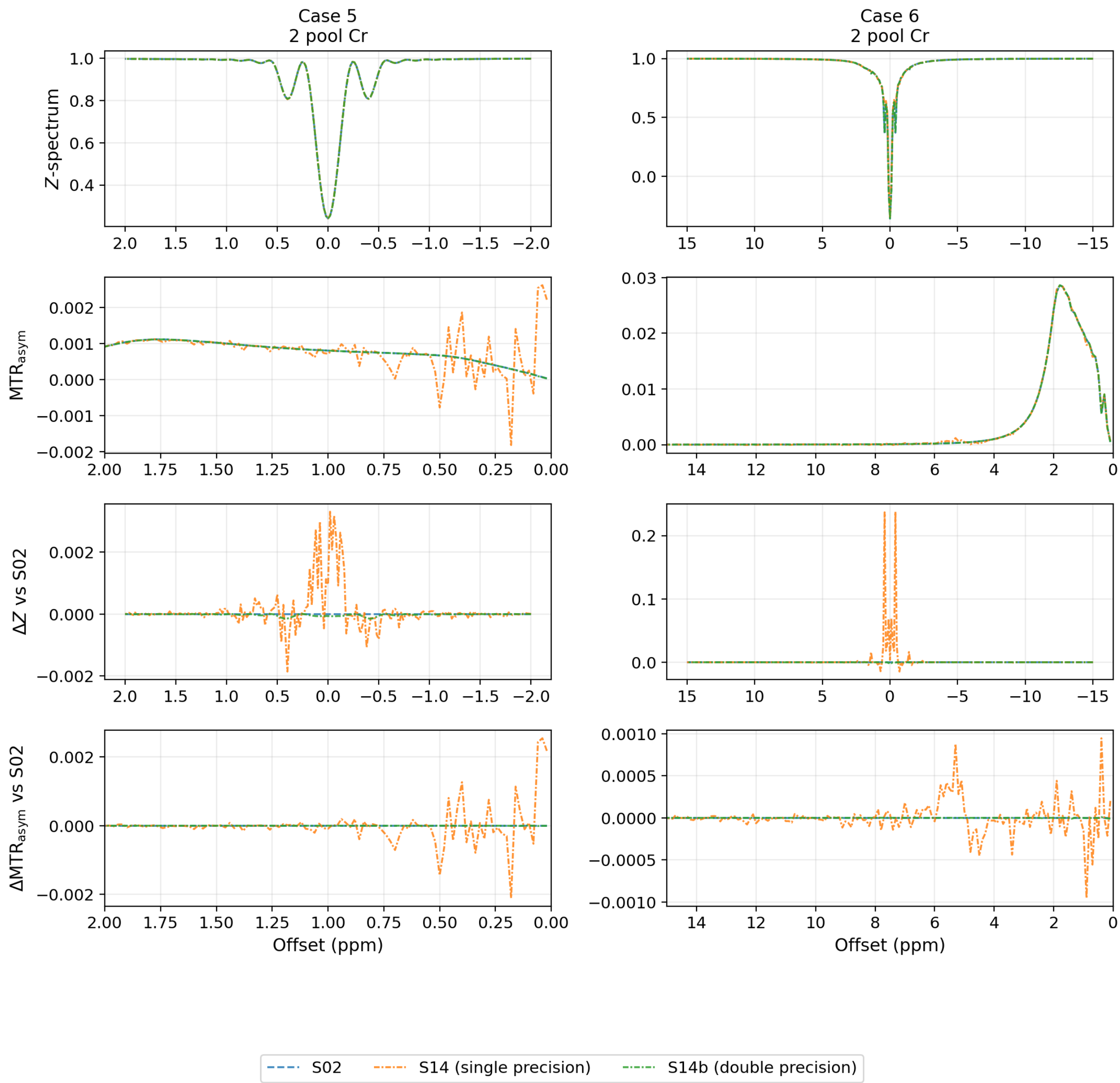

Figure 8: Single- and double-precision BART simulations (S14 and S14b) compared with the reference S02 for Cases 5 and 6. Single-precision BART deviates strongly from S02 in the pulse-train case, whereas double precision agrees much more closely.

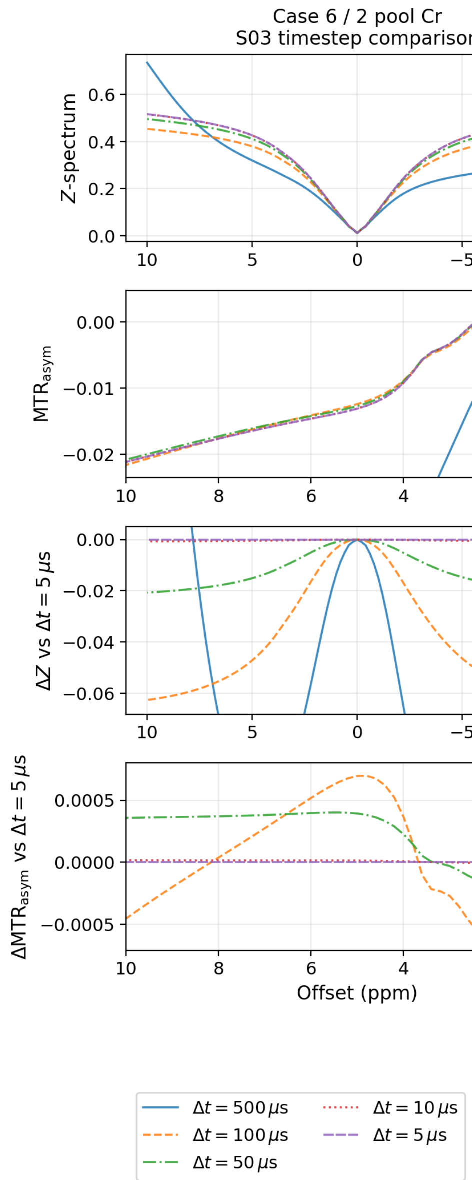

Figure 9: Case 6 Z- and MTR_{asym} spectra for submission S03 simulated with fixed timesteps from 500 to 5 μs. The lower two panels show differences to the finest timestep (5 μs). Axis limits in the lower two panels were chosen to highlight agreement among the smaller timesteps.